FINITE FLAT QUASI-COHERENT MODULES ARE FINITE LOCALLY FREE EXACTLY WHEN EVERY CLOSED SUBSET CLOSED UNDER GENERALIZATIONS IS OPEN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-finite-flat-is-finite-locally-free is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 5068-5077 and proof span 5079-5151 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof contains an explicit sentence 'The scheme case does not follow directly from the affine case, so we simply repeat the arguments.', which is kept verbatim; the repetition that follows is the counted argument and is not a citation.

Editorial note. The subscripted wedge powers and the displayed direct sum decomposition are the author's own notation and were copied unchanged.

Editorial note. 6 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- Flat closed immersions and the same-chapter characterisation of them, which links flatness of a closed immersion to the failure of its support to be closed under generalization. Invoked twice, at proof lines 5085-5088 and again at 5144-5146. Carries the implication from the module condition to the topological condition and closes the last step of the converse.
- Support of a finite type quasi-coherent module is closed, and the stalk at a point of a finite flat module over a local ring is free, as cited from the exterior-algebra and local-algebra chapters. Cited at proof lines 5094-5096 and 5097-5099. Identifies the support, shows it is closed under generalization when the module is flat, and provides the local freeness used in the final Nakayama step.
- Exterior powers of a finite flat quasi-coherent module are again finite flat quasi-coherent, their supports are nested, and the fibre of the i-th exterior power at a point is the i-th exterior power of the fibre. Used at proof lines

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5112-5130. Produces the open and closed decomposition of X indexed by the rank and identifies the rank at each point.
- Nakayama's lemma in the form that lifts a basis of the fibre to a surjection from a free module over an affine neighbourhood. Cited at proof lines 5130-5134. Produces the local surjection O_U to the power r to F restricted to U whose injectivity is the last thing to be proved.

Source reference. Original reference algebra-lemma-NAK: sources/stacks-morphisms/pinned-original-source.tex, source lines 709–709. Same-chapter reference used in the retained author text at source line(s) 709, 5128, 7341, 7915, 11427, 15408. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original reference algebra-lemma-finite-flat-local: sources/stacks-morphisms/pinned-original-source.tex, source lines 5099–5099. Same-chapter reference used in the retained author text at source line(s) 5099. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original reference algebra-lemma-finite-flat-module-finitely-presented: sources/stacks-morphisms/pinned-original-source.tex, source lines 5081–5081. Same-chapter reference used in the retained author text at source line(s) 5081. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma lemma-characterize-flat-closed-immersions: sources/stacks-morphisms/pinned-original-source.tex, source lines 5003–5003. Same-chapter reference used in the retained author text at source line(s) 5061, 5091, 5146. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original reference modules-lemma-support-finite-type-closed: sources/stacks-morphisms/pinned-original-source.tex, source lines 700–700. Same-chapter reference used in the retained author text at source line(s) 700, 5097, 7918. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original reference modules-section-symmetric-exterior: sources/stacks-morphisms/pinned-original-source.tex, source lines 5112–5112. Same-chapter reference used in the retained author text at source line(s) 5112. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let X be a scheme. The following are equivalent*

(1) *every finite flat quasi-coherent $\mathcal{O}_X$-module is finite locally free, and*
(2) *every closed subset $Z \subset X$ which is closed under generalizations is open.*

Proof. In the affine case this is Algebra, Lemma algebra-lemma-finite-flat-module-finitely-presented. The scheme case does not follow directly from the affine case, so we simply repeat the arguments.

Assume (1). Consider a closed immersion $i : Z \to X$ such that i is flat. Then $i_*\mathcal{O}_Z$ is quasi-coherent and flat, hence finite locally free by (1). Thus $Z = \mathrm{Supp}(i_*\mathcal{O}_Z)$ is also open and we see that (2) holds. Hence the implication (1) $\Rightarrow$ (2) follows from the characterization of flat closed immersions in Lemma lemma-characterize-flat-closed-immersions.

For the converse assume that X satisfies (2). Let $\mathcal{F}$ be a finite flat quasi-coherent $\mathcal{O}_X$-module. The support $Z = \mathrm{Supp}(\mathcal{F})$ of $\mathcal{F}$ is closed, see Modules, Lemma modules-lemma-support-finite-type-closed. On the other hand, if $x \rightsquigarrow x'$ is a specialization, then by Algebra, Lemma algebra-lemma-finite-flat-local the module $\mathcal{F}_{x'}$ is free over $\mathcal{O}_{X,x'}$, and

$$\mathcal{F}_x = \mathcal{F}_{x'} \otimes_{\mathcal{O}_{X,x'}} \mathcal{O}_{X,x}.$$

Hence $x' \in \mathrm{Supp}(\mathcal{F}) \Rightarrow x \in \mathrm{Supp}(\mathcal{F})$, in other words, the support is closed under generalization. As X satisfies (2) we see that the support of $\mathcal{F}$ is open and closed. The modules $\wedge^i(\mathcal{F})$, $i = 1, 2, 3, \ldots$ are finite flat quasi-coherent $\mathcal{O}_X$-modules also, see Modules, Section modules-section-symmetric-exterior. Note that $\mathrm{Supp}(\wedge^{i+1}(\mathcal{F})) \subset \mathrm{Supp}(\wedge^i(\mathcal{F}))$. Thus we see that there exists a decomposition

$$X = U_0 \amalg U_1 \amalg U_2 \amalg \ldots$$

by open and closed subsets such that the support of $\wedge^i(\mathcal{F})$ is $U_i \cup U_{i+1} \cup \ldots$ for all i. Let x be a point of X, and say $x \in U_r$. Note that $\wedge^i(\mathcal{F})_x \otimes \kappa(x) = \wedge^i(\mathcal{F}_x \otimes \kappa(x))$. Hence, $x \in U_r$ implies that $\mathcal{F}_x \otimes \kappa(x)$ is a vector space of dimension r. By Nakayama's lemma, see Algebra, Lemma algebra-lemma-NAK we can choose an affine open neighbourhood $U \subset U_r \subset X$ of x and sections $s_1, \ldots, s_r \in \mathcal{F}(U)$ such that the induced map

$$\mathcal{O}_U^{\oplus r} \longrightarrow \mathcal{F}|_U, \quad (f_1, \ldots, f_r) \longmapsto \sum f_i s_i$$

is surjective. This means that $\wedge^r(\mathcal{F}|_U)$ is a finite flat quasi-coherent $\mathcal{O}_U$-module whose support is all of U. By the above it is generated by a single element, namely $s_1 \wedge \ldots \wedge s_r$. Hence $\wedge^r(\mathcal{F}|_U) \cong \mathcal{O}_U/\mathcal{I}$ for some quasi-coherent sheaf of ideals $\mathcal{I}$ such that $\mathcal{O}_U/\mathcal{I}$ is flat over $\mathcal{O}_U$ and such that $V(\mathcal{I}) = U$. It follows that $\mathcal{I} = 0$ by applying Lemma lemma-characterize-flat-closed-immersions. Thus $s_1 \wedge \ldots \wedge s_r$ is a basis for $\wedge^r(\mathcal{F}|_U)$ and it follows that the displayed map is injective as well as surjective. This proves that $\mathcal{F}$ is finite locally free as desired. $\square$